

Operations

Answer Booklet

Past HSC and trial questions from your Drive, screenshotted exactly as they appeared in the original papers.

SECTION	QUESTIONS			
Section A — Multiple choice	Q1–18	18 questions	18 marks	p. 3
Section B — Short answer	Q19–45	27 questions	89 marks	p. 4
Section C — 20-mark questions	Q46–47	2 questions	40 marks	p. 19

How to use this booklet

- Question numbers match the Question Booklet exactly. Each question header shows the syllabus dot point(s) it tests, and the **Syllabus map** (next page) lists every dot point and the questions that cover it.
- ■ **Marking guidelines** – screenshots of the official NESA or school marking guidelines, criteria and sample answers.
- 📄 **Model answer** (yellow box) – written for this booklet wherever the paper had no answer, following the Business Studies structure (define → explain the link → apply to the business/case study).
- Mark yourself against the criteria first, then compare your answer with the sample.

Syllabus map

Every dot point in the syllabus (as in your notes) and the Question Booklet questions that test it. Tick each one off once you can answer its questions confidently.

CODE	SYLLABUS DOT POINT	QUESTIONS	✓
O1a	Strategic role of operations management – cost leadership, good/service differentiation	Q1, Q19, Q45	☐
O1b	Goods and/or services in different industries	Q2	☐
O1c	Interdependence (operations) with other key business functions	Q10, Q20	☐
O2a	Influences: globalisation, technology, quality expectations, cost-based competition, government policies, legal regulation, environmental sustainability	Q3, Q21, Q22, Q23, Q46	☐
O2b	Corporate social responsibility – legal compliance vs ethical responsibility; environmental sustainability and social responsibility	Q3, Q4, Q23, Q24, Q25, Q34	☐
O3a	Inputs – transformed resources (materials, information, customers); transforming resources (human resources, facilities)	Q5, Q26, Q27	☐
O3b	Transformation processes – influence of volume, variety, variation in demand, visibility (4 Vs)	Q6, Q28, Q29	☐
O3c	Sequencing and scheduling – Gantt charts, critical path analysis	Q7, Q8, Q30, Q31	☐
O3d	Technology, task design and process layout	Q32, Q33, Q40	☐
O3e	Monitoring, control and improvement	Q9, Q31	☐
O3f	Outputs – customer service, warranties	Q10, Q34	☐
O4a	Performance objectives – quality, speed, dependability, flexibility, customisation, cost	Q11, Q35, Q36, Q40, Q47	☐
O4b	New product or service design and development	Q37	☐
O4c	Supply chain management – logistics, e-commerce, global sourcing	Q12, Q21, Q38, Q44, Q47	☐
O4d	Outsourcing – advantages and disadvantages	Q13, Q39	☐
O4e	Technology – leading edge, established	Q14, Q21, Q40, Q43, Q47	☐
O4f	Inventory management – advantages/disadvantages of holding stock, LIFO, FIFO, JIT	Q15, Q16, Q41, Q42	☐
O4g	Quality management – control, assurance, improvement	Q10, Q17, Q39, Q44, Q47	☐
O4h	Overcoming resistance to change – financial costs, purchasing new equipment, redundancy payments, retraining, reorganising plant layout, inertia	Q43, Q46	☐
O4i	Global factors – global sourcing, economies of scale, scanning and learning, research and development	Q12, Q18, Q38, Q44, Q45	☐

Section A — Multiple choice

Q	ANSWER	WHY	SOURCE
1	A	Cutting input costs to compete on price = cost leadership	2019 HSC · Q11
2	A	Goods are tangible; services are intangible	Roseville Trial 2023 · Q8
3	D	Environmental protection laws regulate the use and disposal of chemicals in production	2022 HSC · Q10
4	C	Adding healthy menu options changes the operations process for social responsibility; donations are not operations and workers comp is legal compliance	2021 HSC · Q12 worked out
5	D	Transformed resources are the materials changed in the process (burgers, drinks); the rest are transforming resources	Newington Trial 2023 · Q5
6	D	Visibility = how much of the transformation process the customer can see	Roseville Trial 2024 · Q9
7	D	D needs both the C path and the E-F path finished, plus A and B	2019 HSC · Q19
8	C	Critical path A-B-C-D-G becomes $1 + 3 + 1 + 1 = 6$ days	2019 HSC · Q20
9	C	Comparing results with targets and making corrective changes = controlling	Abbotsleigh Trial 2024 · Q9
10	C	A longer warranty is only viable if quality management keeps defects and warranty claims low	2021 HSC · Q14 worked out
11	A	Designing each display to individual couples' preferences is customisation	2023 HSC · Q2 worked out
12	C	Global sourcing obtains cheaper inputs from lower-cost overseas suppliers	2022 HSC · Q7
13	D	Outsourcing brings efficiency/cost savings but can create communication and language barriers	Abbotsleigh Trial 2024 · Q13
14	A	Leading-edge technology can raise output quality but needs high initial investment; 'proven' and 'widely used' describe established technology	2021 HSC · Q3 worked out
15	A	Newest stock treated as sold first = last-in-first-out (LIFO)	Abbotsleigh Trial 2024 · Q5
16	A	JIT holds little stock (lower holding costs) but needs frequent deliveries (higher logistics) and risks stock-outs	Knox Trial 2023 · Q17
17	A	Inspecting to detect defects during production = quality control	2019 HSC · Q9
18	C	Studying global trends and best practice before designing = scanning and learning	2020 HSC · Q13

"worked out" = the source paper had no answer key, so the answer was worked out for this booklet.

Section B — Short answer

1. Role of operations management

Q19 2020 HSC · Question 23 (a) O1a · Strategic role of operations

4 marks

■ MARKING GUIDELINES — 2020 HSC

Question 23 (a)

Criteria	Marks
• Provides detailed ways in which the strategic role of operations management could assist this business	4
• Provides ways in which the strategic role of operations management could assist this business	3
• Provides characteristics and features of operations management that could assist this business	2
• Sketches in general terms an aspect of operations management	1

Sample answer:

The strategic role of operations management ensures the operations function effectively contributes to the achievement of goals and the strategic direction of the business. The business could examine goods/service differentiation as currently the business' product range is limited to denim products. They could differentiate their products and add non-denim items and clothing accessories. This would potentially increase the customer base, attracting customers who are interested in non-denim items which may increase sales and profits.

Q20 Abbotsleigh Trial 2024 · Question 23 (b) O1c · Interdependence M1b · Interdependence

3 marks

■ MARKING GUIDELINES — ABBOTSLEIGH TRIAL 2024

b) Explain the interdependence of operations and marketing for this business.

(3 marks)

Marking Criteria	Mark
Clearly provides why and how operations and marketing are interdependent for this business	3
Provides features and characteristics of the interdependence of operations and marketing for this business	2
Provides some relevant information	1

Suggested response:

Interdependence is the mutual reliance of one function on another in order to meet the goals of the business. For this business, operations will depend on marketing to conduct market research to help with the design of shaving kits that consumers desire. Marketing will depend on operations to produce the required quantities of razors in an appropriate timeframe and to make sure that the razors meets the quality claims made by the marketing team.

2. Influences on operations management

Q21 2020 HSC · Question 23 (b)

O2a · Operations influences

O4e · Technology

O4c · Supply chain management

6 marks

■ MARKING GUIDELINES — 2020 HSC

Question 23 (b)

Criteria	Marks
• Draws out and relates the implications of using operations strategies to address TWO operations influences affecting the business	6
• Provides implications of operations strategies that could be used to address TWO operations influences affecting the business	5
• Provides characteristics and features of operations strategies which could be used by this business to address TWO operations influences affecting the business	4
• Sketches in general terms operations strategies in general terms with limited link to operations influences	3
• Sketches in general terms a strategy or an operations influence in general terms	2
• Provides some relevant information	1

Sample answer:

The business should address the influence of technology. To manage this influence, the store could adopt more leading-edge payment methods such as smartphones and smart watches or established technologies such as EFTPOS. By purchasing this new equipment, customers might be more inclined to shop at the business which may increase sales and enhance speed of service.

To address the influence of cost-based competition, which has resulted in declining sales, the business could utilise the strategies of supply chain management, global sourcing or the performance objective of cost. By improving logistics in the supply chain and focusing on cost as a performance objective the business could improve efficiency and better compete with foreign businesses. Moreover, global sourcing typically enables lower input costs as the business takes advantage of low cost clothing producers.

■ MARKING GUIDELINES — NEWINGTON TRIAL 2023

(c) Outline how quality expectations could impact this business.

Criteria	Mark
▪ Provides main characteristics and features of how quality expectations could impact this business using relevant terminology, syllabus concepts and information from the stimulus	2
▪ Provides a characteristic or feature of how quality expectations may impact the business and makes limited/no reference to stimulus, OR ▪ Defines quality expectations	1
▪ No response or factually incorrect response	0

Sample answer:

- Customers have a pre-existing idea about the quality of a certain product or brand. Beliefs about durability, reliability, fitness for purpose – operations need to be organised to maximise customer satisfaction.
- Operations needs to match the customers' expectations. If the business falls short of customer expectations, it may suffer long-term damage to its goodwill and reputation in the market.
- Refer to the stimulus/business.

Quality expectations are the perceptions surrounding a business's product from consumers. Rebecca's health food business could be damaged by quality expectations as the damaged goods may give consumers the perception of lower quality, combined with the increased competition from retail chains, they may not justify the prices of her products with the perceived value, leading to a decrease in sales.

Quality expectations, expectations surrounding the quality of a product, could negatively impact this business as consumers will begin to form an idea and expectation surrounding Rebecca's products as damaged and unavailable ("out of stock"). Thus, if this doesn't improve, her quality expectations will decrease her sales and reputation, but if she does improve, the opposite will occur.

■ MARKING GUIDELINES — SIC TRIAL 2024

(b) A market leading business manufactures electronics.

4

Explain how environmental sustainability can influence the operations of this business.

<i>Criteria</i>	<i>Mark</i>
Clearly provides an explanation of how environmental sustainability can influence the operations of this business	4
Provides an explanation of how environmental sustainability can influence the operations of this business	3
Sketches in general terms how environmental sustainability can influence the operations of this business	2
Makes a relevant statement about environmental sustainability	1

Answers could include:

- Increase costs of compliance to reach standards of environmental sustainability
- Increase complexity of operations to meet environmental standards
- Change of inputs to be more environmentally friendly
- Limiting waste which can alter processes and increase costs
- All of this leads to customer satisfaction, which can increase sales and market share.

■ MARKING GUIDELINES — BARKER TRIAL 2024

(a) Distinguish between legal compliance and corporate social responsibility.

Criteria	Marks
Correctly recognises the differences between legal compliance and CSR	2
Sketches aspects of legal compliance or CSR	1

Sample answer:

Legal compliance involves adhering to the relevant laws, regulations and rules set by government bodies and is therefore mandatory and enforced by law. On the other hand, corporate social responsibility are voluntary actions taken by a business to act ethically and contribute positively to society which is beyond their legal requirements.

■ MARKING GUIDELINES — 2019 HSC

Question 24 (c)

Criteria	Marks
• Provides detailed reasons why this business might be reluctant to become more socially responsible	4
• Provides reasons why this business might be reluctant to become more socially responsible	3
• Sketches in general terms how social responsibility may affect this business	2
• Provides some information about social responsibility	1

Sample answer:

This business may be reluctant in meeting its social responsibility due to the increased cost in purchasing environmentally friendly products and packaging such as biodegradable straws and cups.

Additionally, the business will need to find a new supplier for the environmentally friendly straws and cups. The cost of the new packaging may lead to an increase in the price of juice for the customers, possibly resulting in a decrease in sales.

3. Operations processes

■ MARKING GUIDELINES — 2019 HSC

Question 24 (a)

Criteria	Marks
• Identifies TWO transforming resources	2
• Identifies ONE transforming resource	1

Sample answer:

Human Resources, Facilities

MODEL ANSWER

Inputs are the resources used in the operations process, and **customers are a transformed resource**. This customer is an input because they supply the information the whole build is based on – the house design, specifications, fittings, budget and the six-month deadline written into the contract – and they stay involved during construction by approving plans and choosing finishes. The builder processes these requirements and "transforms" them into the finished home that satisfies the customer.

■ MARKING GUIDELINES — 2019 HSC

Question 24 (b)

Criteria	Marks
• Shows in detail how variety impacts the transformation process for this business	4
• Shows how variety impacts the transformation process for this business	3
• Sketches in general terms the transformation process	2
• Provides some information about the transformation process	1

Sample answer:

Variety refers to the range of products made by the business. With a wide variety of juice being served the business is able to offer greater choice to customers. The variation in the transformation process may include new juice machinery, systems and procedures. This will cost the business money in terms of training and resources, therefore impacting profitability. However, this will enable the business to have a wider target market, leading to more customers, greater sales and improved profitability.

■ MARKING GUIDELINES — REDDAM TRIAL 2023

- (d) Discuss the impact of variety and visibility on the transformation process of the car retailer.

Outcome: H4.

Criteria	Marks
Identify issues and provides points for and against the impact of variety and visibility on the transformation process of the car retailer.	3 - 4
Refers to either points for or against the impact of variety and visibility on the transformation process of the car retailer.	1 - 2

Answer:

- **Variety:** greater variety will enable the car producer to recruit a wider range of customers increasing sales and profits. Variety also spreads out the risk of the car retailer if one car type or model loses popularity. However, stocking a wider variety of cars will increase inventory costs potentially decreasing profits. Sales staff will also require additional training to gain sufficient product knowledge of all car types.
- **Visibility or customer contact:** improves the transformation process because customer feedback can be used to improve the quality, price or range of cars stocked by the car retailer, increasing their competitiveness and profits. Visibility also increases the cost of transformation processes because the car retailer may need to invest more money in equipment and/or staff to obtain customer feedback.

MODEL ANSWER

A **Gantt chart** is a bar chart that shows every task in a project against a timeline – when each task starts and finishes, how long it takes, and which tasks must be done in **sequence** and which can run **at the same time**.

Meeting the six-month deadline: the manager can sequence the build in the correct order (slab → frame → roof → plumbing and electrical → fit-out) and schedule each trade to arrive exactly when the previous stage is finished, so there is no idle time. Tasks that do not depend on each other (e.g. landscaping and internal painting) can be scheduled concurrently, shortening the total project time.

Staying within budget: because actual progress can be monitored against the plan, delays are spotted early and resources can be reallocated before they become costly (overtime, extra equipment hire, penalty clauses). Materials and subcontractors are booked only when needed, reducing wasted labour and storage costs.

The Gantt chart is therefore both a planning and a control tool that helps the builder deliver the home on time and on budget.

MODEL ANSWER

Gantt charts and critical path analysis (CPA) are sequencing and scheduling tools that plan the order and timing of tasks.

- A **Gantt chart** is a bar chart showing each task against a timeline, including when it starts and finishes and which tasks overlap. Managers can see at a glance whether a project is on schedule, allocate staff and equipment to the right place at the right time, and **monitor progress against the plan**, taking corrective action early if tasks fall behind.
- **Critical path analysis** uses a network diagram to identify the **critical path** – the longest sequence of dependent tasks, which determines the minimum time to complete the project. Tasks on the critical path cannot be delayed without delaying the whole project, so managers can focus resources on them, while tasks with **slack (float)** can be scheduled flexibly.

Together they improve operations by **reducing idle time and bottlenecks, shortening completion times, lowering costs** (less overtime and wasted resources) and helping the business meet deadlines – improving the performance objectives of **speed, dependability and cost**, and supporting monitoring, control and improvement of the transformation process.

■ MARKING GUIDELINES — 2022 HSC

Question 21 (c)

Criteria	Marks
• Provides detailed similarities and/or differences about the efficiency and profitability of each layout	4
• Provides some similarities and/or differences about efficiency and profitability of each layout	3
• Provides similarities and/or differences about efficiency and/or profitability of the layouts	2
• Provides relevant information about process layout	1

Sample answer:

The layout of option 2 allows employees to make a finished burger efficiently as the process to make burgers is sequential. In comparison, option 1's narrow kitchen layout requires employees to go back and forth between work stations to make the burger which could lead to accidents and time lost, causing it to be more inefficient. Option 1 has more tables which could seat more customers and therefore increase profitability compared to option 2 which has fewer tables for dine-in customers, therefore it is unable to maximise profitability.

■ MARKING GUIDELINES — BARKER TRIAL 2023

(c) Outline the purpose of task design.

Criteria	Marks
• Sketches in general terms the purpose of task design in operations	2
• Makes an accurate statement regarding task design	1
• No valid response	0

Marker's comments:

Task design involves classifying job activities in ways that make it easier for an employee to successfully perform and complete tasks in the operations process. It involves breaking down the tasks required to create a job analysis and job description.

Weaker responses were unable to provide a definition of the concept of task design but may have correctly stated the purpose or benefit.

■ MARKING GUIDELINES — REDDAM TRIAL 2023

- (c) Why may this business exceed its legal obligations when offering warranties to customers?

Outcome: H2.

Criteria	Marks
Provides cause and effect of why this business may exceed its legal obligations when offering warranties to customers.	2
Sketches in general terms why this business may exceed its legal obligations when offering warranties to customers.	1

Answer:

- Longer warranties reinforce the car retailer's reputation of selling reliable cars, enabling the car retailer to obtain a competitive advantage increasing sales and profits. Longer warranties will enhance the brand image of the car retailer, increasing sales and profits. Longer warranties also enable the business to charge a higher price for their cars, maximising their profits.

4. Operations strategies

■ MARKING GUIDELINES — 2022 HSC

Question 21 (a)

Criteria	Marks
• Sketches in general terms a performance objective relevant to the business	2
• Identifies an objective for this business	1

Sample answer:

- Dependability would be a key performance objective for this business as customers expect consistency. They want their burgers to be served in the same timeframe, be the same size, temperature and taste each time they purchase one.

Answers could include:

- Quality, speed, dependability, flexibility, customisation, cost.

■ MARKING GUIDELINES — REDDAM TRIAL 2024

(b) Explain ONE possible conflict that may arise between achieving TWO performance objectives for the sports shoe manufacturer.

Outcome: H5.

Criteria	Marks
Provides cause and effect of ONE possible conflict that may arise between achieving TWO performance objectives for the sports shoe manufacturer.	2
Provides characteristics and features of ONE possible conflict that may arise between achieving TWO performance objectives for the sports shoe manufacturer.	1

Suggested Answers:

- Dependability/quality may compromise (increase) cost and vice versa, quality and customisation may compromise (lengthen lead times) speed, speed may conflict with quality.
-

■ MARKING GUIDELINES — NEWINGTON TRIAL 2023

- (b) A business has decided to expand its product range. Discuss the operational strategy of new product or service design and development for this business.

Criteria	Mark
Clearly provides four points for and/or against new product or service design and development on the business using relevant terminology and syllabus concepts.	4
- Clearly provides at least three points for and/or against new product or service design and development on the business, OR - Provides detailed points for and/or against new product or service design and development.	3
- Provides at least one detailed point for or against new product or service design and development, OR - Clearly provides two points for and/or against new product or service design and development, OR	2
- Provides a characteristic or feature of new product or service design or development, OR - Provides one point for or against new product or service design and development	1
No response or factually incorrect response	0

Sample answer:

Advantages for the business could include:

- An increase in sales
- Increase in market share
- Increase in profits

Disadvantages for the business could include:

- Could be expensive to expand its product range
- May be time consuming implementing new products

New product or service design and development can increase the target market for the business. By appealing to more customers, the business will be able to service the broader community, ultimately bringing in new customers, resulting in greater profits. However, the expansion of a business' product range may result in the detracting from previous operations. It may require the purchase of additional equipment, the changing of the plant layout or hiring of new staff which ultimately increases the short term operational costs.

■ MARKING GUIDELINES — ROSEVILLE TRIAL 2024

- (d) Explain how the operations strategies of global sourcing and economies of scale might be beneficial to this business.

4

<i>Criteria</i>	<i>Mark</i>
Provides a detailed cause and effect relationship as to how the concepts of global sourcing and economies of scale might be beneficial to this business.	4
Provides a cause and effect relationship as to how the concepts of global sourcing and economies of scale might be beneficial to this business.	3
Sketches in general terms how the concepts of global sourcing might be beneficial.	2
Makes a relevant statement about global sourcing or economies of scale.	1

Answers could include:

- Definitions of global sourcing and economies of scale
- Global sourcing in relation to specialty raw material inputs (sushi, rice, other food inputs) as well as technology (conveyor belt, specialist sushi equipment or labour (key staff or personal to cook and train chefs here)).
- Economies of scale – increased global footprint allows for bulk purchases, savings that come from buying in bulk, reduced costs and competitive advantages.

📌 MODEL ANSWER

1. Quality control – inspecting work against set standards to find and fix defects. The operations manager could inspect each outsourced stage (e.g. framing, plumbing, electrical) at key milestones and only sign it off – and pay the subcontractor – once it meets the building standards and the customer's specifications. Faults are corrected before later stages cover them up, avoiding costly rework and delays to the six-month deadline.

2. Quality assurance – a preventative system that makes sure standards are met before and during production. The builder could only outsource to licensed tradespeople or firms with a certified quality system (e.g. ISO 9001) and write the required standards, materials and completion dates into each subcontract, with penalty clauses for non-compliance. Quality is built in from the start, reducing the risk of defective outsourced work and protecting the builder's reputation and its ability to meet the contract.

■ MARKING GUIDELINES — BARKER TRIAL 2024

(c) Analyse the implementation of technology in achieving operations objectives for this business.

Criteria	Marks
Draws out and relates the implications of the implementation of technology in achieving the business's operations objectives	5
Shows how/why the implementation of technology achieves the business' operations objectives	4
Provides characteristics and features of the implementation of technology in achieving operations objectives	3
Sketches in general terms how implementing technology achieves objectives	2
Sketches aspects of technology or business objectives	1

Sample answer:

Technology refers to the development and application of innovative designs, methods and devices that act as a transforming resource in the transformation process. By using leading edge technology such as robotics as part of an automated assembly line Focus Frames is able to complement or substitute their workers with machinery, reducing labour costs. Efficient and innovative technologies will perform tasks faster than staff, with greater accuracy and precision, improving dependability. Although technology will have a high initial cost, it will improve economies of scale and increase output, lowering cost and improving speed, contributing to a competitive advantage for FF.

■ MARKING GUIDELINES — 2022 HSC

Question 21 (b)

Criteria	Marks
Provides reasons in favour of an appropriate inventory management strategy for the business	3
Provides a reason in favour of an appropriate inventory management strategy for the business	2
Makes a relevant statement about inventory management	1

Sample answer:

Just-in-time would be an effective inventory method as it would allow the business to take advantage of fresh produce as it arrives. This would also save them storage space and the costs associated with this.

Answers could include:

- Last in first out
- First in first out.

■ MARKING GUIDELINES — REDDAM TRIAL 2023

(b) Describe ONE disadvantage of holding stock for the car retailer.

Outcome: H5.

Criteria	Marks
Provides characteristics and features of ONE disadvantage of holding stock for the car retailer.	2
Sketches in general terms ONE disadvantage of holding stock for the car retailer.	1

Answer:

- **Disadvantages:** older cars will become obsolete when the latest model is released therefore the car retailer may need to decrease price eroding their profit margin. Cars are bulky and need to be stored therefore storage, insurance and spoilage costs increases.

🔑 MODEL ANSWER

Introducing leading-edge technology is a major change, and both staff and management may resist it.

Staff may resist because of: - **Fear of job loss (redundancy)** – automated equipment may replace manual jobs, creating insecurity and lowering morale. - **Fear of the unknown / lack of skills** – employees may worry they cannot learn the new technology and must undergo **retraining**, disrupting familiar routines. Changes to job design and the **plant layout** may alter how and where they work. - **Inertia** – people are comfortable with established ways of working and often resist change unless its benefits are explained.

Management may resist because of: - **Financial costs** – the high cost of purchasing new equipment, installation, **redundancy payments** and **retraining** reduces short-term profit and cash flow, and there is a risk the investment does not deliver the expected economies of scale. - **Disruption and risk** – production may slow during installation and while staff learn the system; leading-edge technology is untested and may have faults or quickly become obsolete. - **Inertia and loss of control** – managers may prefer established systems they understand and fear conflict with staff and unions.

To overcome resistance the business can communicate the reasons for the change, consult and involve employees, provide training and support, and introduce the technology gradually.

MODEL ANSWER

One issue is that **cheaper global rubber may be of lower or inconsistent quality**. Global sourcing reduces the cost of inputs, but suppliers in other countries may use different quality standards, and the business has less control over them. If lower-grade rubber is used, the tyres may be less durable or unsafe, leading to defects, warranty claims and recalls. The business may then need extra quality control (inspections, testing, supplier audits), which **adds costs** and reduces the savings it hoped to make – so it must balance lower input costs against maintaining quality outputs.

MODEL ANSWER

Economies of scale are the cost savings a business gains as it increases its scale of production, because fixed costs are spread over more units and inputs such as rubber can be bought in bulk at a discount, lowering the **average cost per tyre**.

- **Benefits for the business:** lower cost per unit increases **profit margins** and supports a **cost leadership** strategy. The business can price more competitively, increasing sales and market share, and the extra profit can be reinvested (e.g. in new technology or R&D).
- **Benefits for consumers:** the savings can be passed on as **lower prices** for tyres of the same quality, and higher output improves **availability**, so consumers get better value for money.

Section C — 20-mark questions

Q46

2021 HSC · Question 25

O2a · Operations influences

O4h · Resistance to change

F2b · External finance

20 marks

MODEL RESPONSE PLAN

Business report format: use headings, an introduction and recommendations. Address every dot point in the question.

To: Owners, Ozzi Baby Food Pty Ltd **From:** Business Consultant **Re:** Proposed automated factory

Introduction

Ozzi Baby Food's safe, sustainably sourced products have created strong domestic and global demand. This report outlines two operations influences on the business, discusses how retraining and redundancy payments can overcome employee resistance to automation, and compares a mortgage with a new issue of shares to fund the factory.

1. Operations influences

- **Quality expectations** – baby food must meet strict food safety and labelling standards, and parents expect consistently safe, high-quality products. Ozzi's reputation depends on this, so the new automated factory must include rigorous quality control and assurance (e.g. HACCP, ISO certification).
- **Environmental sustainability / CSR** – Ozzi's "sustainably-sourced" range means customers expect ethical sourcing and low-impact production. The factory should use energy-efficient equipment and minimise waste; failing to do so would damage the brand. (Other valid influences: globalisation – access to global markets and competitors; technology – automation increases volume and consistency; legal regulation – food standards.)

2. Overcoming employee resistance – retraining and redundancy payments

Automation will lead to job losses, causing fear, insecurity and inertia among employees. - **Retraining** gives existing employees the skills to operate, monitor and maintain the automated equipment. It reduces fear of job loss, retains experienced staff who understand Ozzi's quality standards and culture, and improves morale. It involves short-term training costs and lost productivity while staff learn. - **Redundancy payments** compensate employees whose roles are no longer needed, as required under the National Employment Standards. Generous payments and outplacement support treat staff fairly, reduce disputes and legal risk, and protect Ozzi's reputation as an ethical employer, but they are a significant financial cost. Used together – retraining where possible and fair redundancy packages where necessary, supported by early consultation – they will minimise resistance and disruption.

3. Mortgage vs new issue of shares

	Mortgage (debt)	New issue of shares (equity)
Ownership/control	Owners keep full control	Must convert to a public company; ownership diluted among new shareholders
Cost	Interest payments (tax deductible); fixed repayments	No interest or repayments, but dividends expected; costly listing (prospectus, ASX fees)
Risk	Increases gearing and financial risk; property is security	Lower gearing and financial risk
Suitability	Long-term loan matched to a long-term asset (the factory)	Raises large amounts of long-term capital; greater reporting obligations (ASIC, ASX)

Recommendations

1. Maintain strict quality management and sustainable practices in the new factory to protect Ozzi's reputation.
2. Consult employees early, retrain as many as possible and offer fair redundancy packages to the remainder.

3. Fund the factory with a **mortgage**: it matches a long-term asset with long-term finance, keeps ownership and control with the current owners, and interest is tax deductible. Ozzi's rising demand should provide the cash flow to meet repayments; gearing should be monitored against industry averages.

Conclusion

With careful quality management, a fair approach to staff and appropriate long-term debt finance, the automated factory will allow Ozzi Baby Food to meet growing demand while protecting its trusted reputation.

■ MARKING GUIDELINES — 2019 HSC

Question 27

Criteria	Marks
• Draws out and clearly relates the implications of the relationship between performance objectives and operations strategies • Applies relevant case study/studies and contemporary business issues • Presents a sustained, logical and cohesive response and communicates clearly using relevant business terminology and concepts	17–20
• Makes evident the relationship between performance objectives and operations strategies • Uses relevant case study/studies and contemporary business issues • Presents a logical and cohesive response using relevant business terminology and concepts	13–16
• Provides characteristics and features of performance objectives or operations strategies AND sketches operations strategies or performance objectives • Makes reference to case study/studies and/or contemporary business issues • Communicates using relevant business terminology and concepts	9–12
• Sketches performance objective(s) and/or operations strategies in general terms • May make reference to case study/studies and/or contemporary business issues • Communicates using some business terminology and concepts	5–8
• Makes limited reference to Operations • May identify case study/studies • Uses basic business terminology	1–4

Answers could include:

Performance objectives – quality, speed, dependability, flexibility, customisation, cost

- New product or service design and development
- Supply chain management – logistics, e-commerce, global sourcing
- Outsourcing – advantages and disadvantages
- Technology – leading edge, established
- Inventory management – advantages and disadvantages of holding stock, LIFO, FIFO, JIT
- Quality management
 - Control
 - Assurance
 - Improvement
- Overcoming resistance to change – financial costs, purchasing new equipment, redundancy payments, retraining, reorganising plant layout, inertia
- Global factors – global sourcing, economies of scale, scanning and learning, research and development.